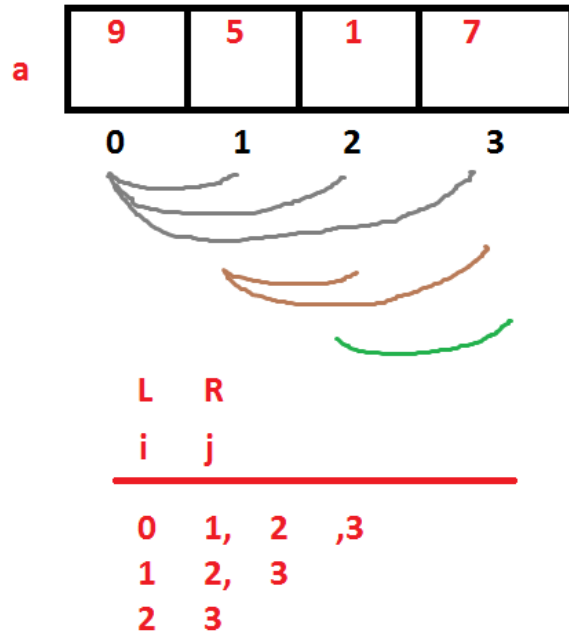


Sorting: Arranging data in ascending or descending order.

Selection sort in ascending order:

9	5	1	7
5	9	1	7
1	9	5	7
1	5	9	7
1	5	7	9



```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 36 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{if(a[i]>a[j]){t=a[i];a[i]=a[j];a[j]=t;}}
}
printf("sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:40 PM
26-Mar-25

```

Enter array size 9
Enter 9 integers 9 2 -4 8 9 0 1 -5 6
Sorted elements are -5 -4 0 1 2 6 8 9 9

```

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```

for( i=0; i<=n-2; i++)
{
    for( j=i+1; j<=n-1; j++)
    {
        L R
        if(a[i] > a[j])
        {
            t=a[i]; a[i]=a[j]; a[j]=t;
        }
    }
}

```

$\frac{n}{4}$ $\frac{i}{0}$ $\frac{j}{1234}$
 1 2 3 4
 2 3 4
 3

a

8 1	4 5	2 7	9 9
0	1	2	3

	L	R
	i	j
✓	0	1, 2, 3
✓	1	2, 3
✓	2	3

Descending order:

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{if(a[i]<a[j]){t=a[i];a[i]=a[j];a[j]=t;}}
}
printf("sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Activate Windows
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F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

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26-Mar-25

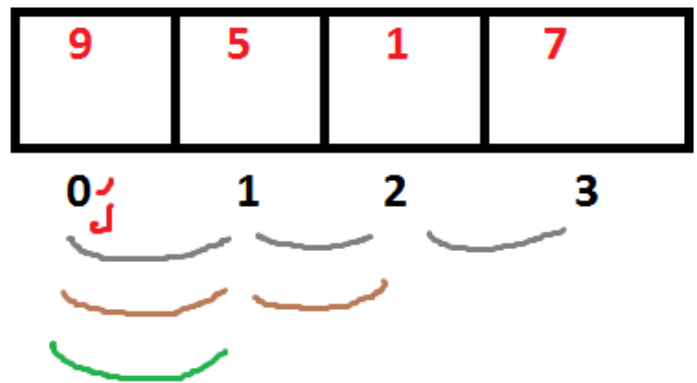
```
TC
Enter array size 9
Enter 9 integers 9 5 1 0 -5 9 -2 8 5
Sorted elements are 9 9 8 5 5 1 0 -2 -5

Activate Windows
Go to PC settings to activate Windows.
```

Bubble sort in ascending order:

9	5	1	7
5	9	1	7
5	1	9	7
5	1	7	9
1	5	7	9

a



L R

i	j, j+1
0	0-1, 1-2, 2-3
1	0-1, 1-2
2	0-1

```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 46 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{if(a[j]>a[j+1]){t=a[j];a[j]=a[j+1];a[j+1]=t;}}
}
printf("sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:15 PM
26-Mar-25


```

Enter array size 9
Enter 9 integers 3 0 -1 9 4 -6 9 1 4
Sorted elements are -6 -1 0 1 3 4 4 9 9

```

Activate Windows
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for(i=0; i<=n-2; i++)

{

4-2-2=0 ✓

for(j=0; j<=n-i-2; j++)

{

L R

if(a[j]>a[j+1])

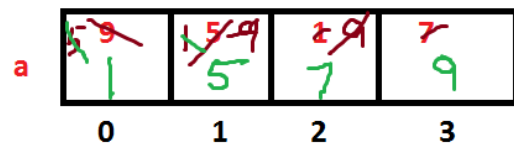
{

t=a[j];a[j]=a[j+1];a[j+1]=t;

}

}

}



	<u>i</u>	<u>j</u>	<u>j+1</u>	<u>L R</u>
	<u>i</u>	<u>j</u>	<u>j+1</u>	<u>i j,j+1</u>
4/n	0	0	1	✓ 0 0-1,1-2,2-3
		1	2	1 0-1,1-2
		2	3	✓ 2 0-1
		3		j<= n-i-2
	1	0	1	4-0-2=2
		1	2	4-1-2=1
3 2		0	1	4-2-2=0

Descending order:

```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{if(a[j]<a[j+1]){t=a[j];a[j]=a[j+1];a[j+1]=t;}}
}
printf("sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:17 PM
26-Mar-25

```
TC
Enter array size 9
Enter 9 integers 9 0 1 -3 -9 -3 8 4 7
Sorted elements are 9 8 7 4 1 0 -3 -3 -9

Activate Windows
Go to PC settings to activate Windows.
```

Find the 2nd max, 2nd min elements of sorted array?

```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,j,t;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<=n-2;i++)
{
for(j=0;j<=n-i-2;j++)
{if(a[j]>a[j+1]){t=a[j];a[j]=a[j+1];a[j+1]=t;}}
}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
for(i=1;i<n;i++)
{if(a[i]>a[0]){printf("\n2nd Min=%d\n",a[i]);break;}}
for(i=n-2;i>=0;i--)
{if(a[i]<a[n-1]){printf("2nd Max=%d",a[i]);break;}}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:54 PM
26-Mar-25

```

Enter array size 9
Enter 9 integers 1 2 3 4 5 6 7 8 9
Sorted elements are 1 2 3 4 5 6 7 8 9
2nd Min=2
2nd Max=8_

```

Activate Windows
Go to PC settings to activate Windows.

1	1	1	6	9	9
0	1	2	3	4	5

```

for(i=1; i<n; i++)
{if(a[i]>a[0])p("2nd min=%d",a[i]);break;}
1 2 3 4 5 6 7 8 9
2
4
for( i=n-2; i>=0; i--)
if(a[i]<a[n-1]){p("2nd max=a[i]);break;}
9 9
6 < 9

```

```

1
1
2
3
4

```

Home work?

1. Finding Nth max, Nth min array elements of sorted array?

-4 -1 0 1 3 6 9 12 70

Eg: 3rd min is 0 and 4th max 6

2. print even elements in ascending order and odd elements in descending order.

Eg: Even -4 0 6 12 70 Odd 9 3 1 -1

3. Arrange half elements in ascending and half elements in descending order

-4 -1 0 1 3 70 12 9 6